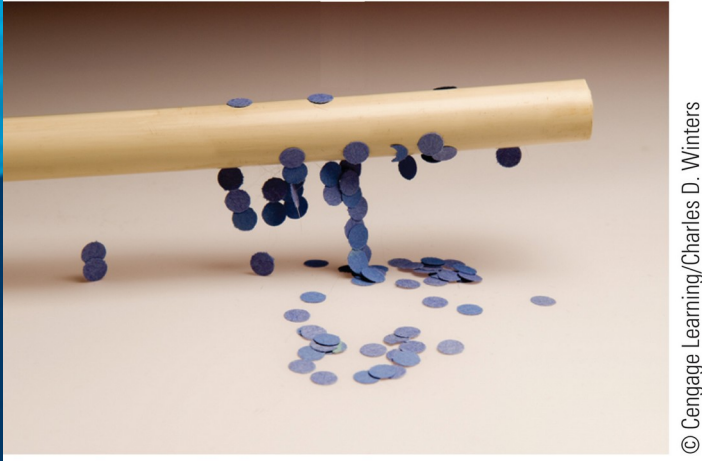




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Topic 15

Electric Forces and Fields

Reading Question 15-1 (1 of 2)

After a glass rod is rubbed against a silk cloth, the glass rod has a charge of $+5$ and the silk cloth has a charge of -7 . What could the original charges of the glass rod and silk cloth have been?

1. $+2, -5$
2. $0, -2$
3. $+5, 0$
4. $0, -5$

Reading Question 15-1 (2 of 2)

After a glass rod is rubbed against a silk cloth, the glass rod has a charge of $+5$ and the silk cloth has a charge of -7 . What could the original charges of the glass rod and silk cloth have been?

1. $+2, -5$
- 2. Answer: $0, -2$**
3. $+5, 0$
4. $0, -5$

Reading Question 15-2 (1 of 2)

You are given four objects, A, B, C, and D. You discover that A and B repel each other, and that C and D repel each other. If B and C attract each other, which of the following are possible charges for A, B, C, and D in that order?

1. A +, B +, C −, D +
2. A −, B +, C +, D −
3. A −, B −, C −, D +
4. A +, B +, C −, D −

Reading Question 15-2 (2 of 2)

You are given four objects, A, B, C, and D. You discover that A and B repel each other, and that C and D repel each other. If B and C attract each other, which of the following are possible charges for A, B, C, and D in that order?

1. A +, B +, C −, D +
2. A −, B +, C +, D −
3. A −, B −, C −, D +
4. **Answer: A +, B +, C −, D −**

Reading Question 15-3 (1 of 2)

A test charge of $-q$ at point P is used to measure an electric field. The field is found to have a magnitude of 2 N/C. When the charge $-q$ is removed, the electric field at point P

1. changes in a way that cannot be determined from the information given.
2. will have magnitude 2 N/C.
3. will reverse in direction.
4. will change in magnitude.

Reading Question 15-3 (2 of 2)

A test charge of $-q$ at point P is used to measure an electric field. The field is found to have a magnitude of 2 N/C . When the charge $-q$ is removed, the electric field at point P

1. changes in a way that cannot be determined from the information given.
- 2. Answer: will have magnitude 2 N/C .**
3. will reverse in direction.
4. will change in magnitude.

Reading Question 15-4 (1 of 2)

Point A is close to an electron. Point B is further from the electron. Points A and B and the electron are along a straight line. Which of the following statements are true?

1. The field is stronger at point A than B and is directed from B to A.
2. The field is weaker at point A than B and is directed from A to B.
3. The field is weaker at point B than A and is directed from A to B.
4. The field is stronger at point B than A and is directed from B to A.

Reading Question 15-4 (2 of 2)

Point A is close to an electron. Point B is further from the electron. Points A and B and the electron are along a straight line. Which of the following statements are true?

- 1. Answer: The field is stronger at point A than B and is directed from B to A.**
2. The field is weaker at point A than B and is directed from A to B.
3. The field is weaker at point B than A and is directed from A to B.
4. The field is stronger at point B than A and is directed from B to A.

Reading Question 15-5 (1 of 2)

Which of the following will change the flux through a gaussian surface caused by a charge distribution?

1. changing the shape of the gaussian surface enclosing the charge distribution
2. changing the size of the gaussian surface enclosing the charge distribution
3. changing the location of the charge distribution within the surface
4. changing the net charge within the gaussian surface

Reading Question 15-5 (2 of 2)

Which of the following will change the flux through a gaussian surface caused by a charge distribution?

1. changing the shape of the gaussian surface enclosing the charge distribution
2. changing the size of the gaussian surface enclosing the charge distribution
3. changing the location of the charge distribution within the surface
4. **Answer: changing the net charge within the gaussian surface**

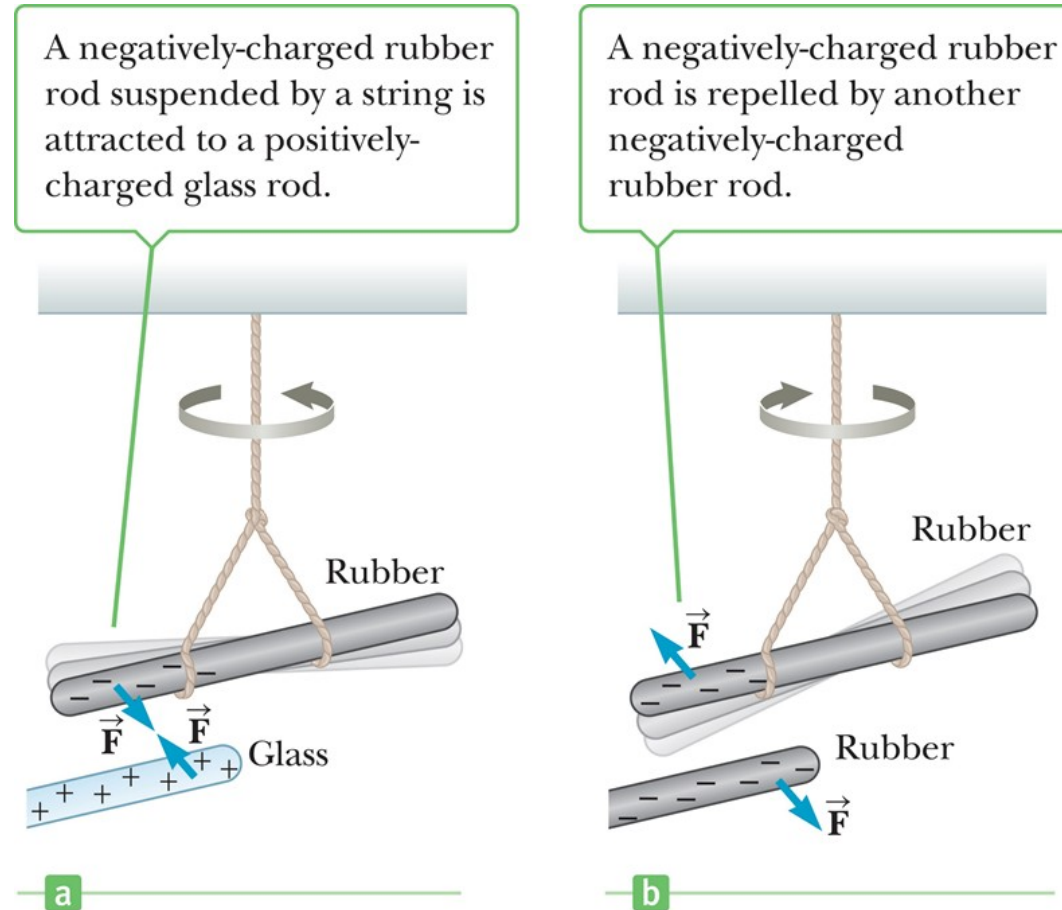
Topic 15: Electric Forces and Fields

TOPIC DISCUSSION

Properties of Electric Charges (1 of 4)



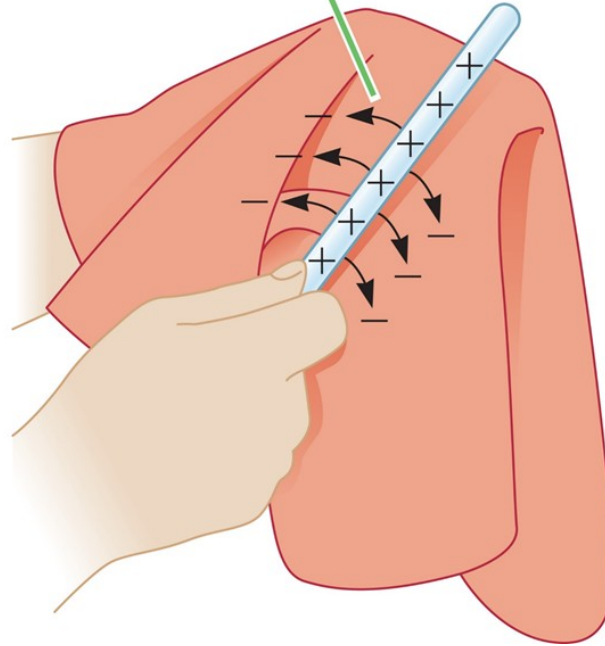
Properties of Electric Charges (2 of 4)



Like charges repel, unlike charges attract.

Properties of Electric Charges (3 of 4)

Each (negatively-charged) electron transferred from the rod to the silk leaves an equal positive charge on the rod.



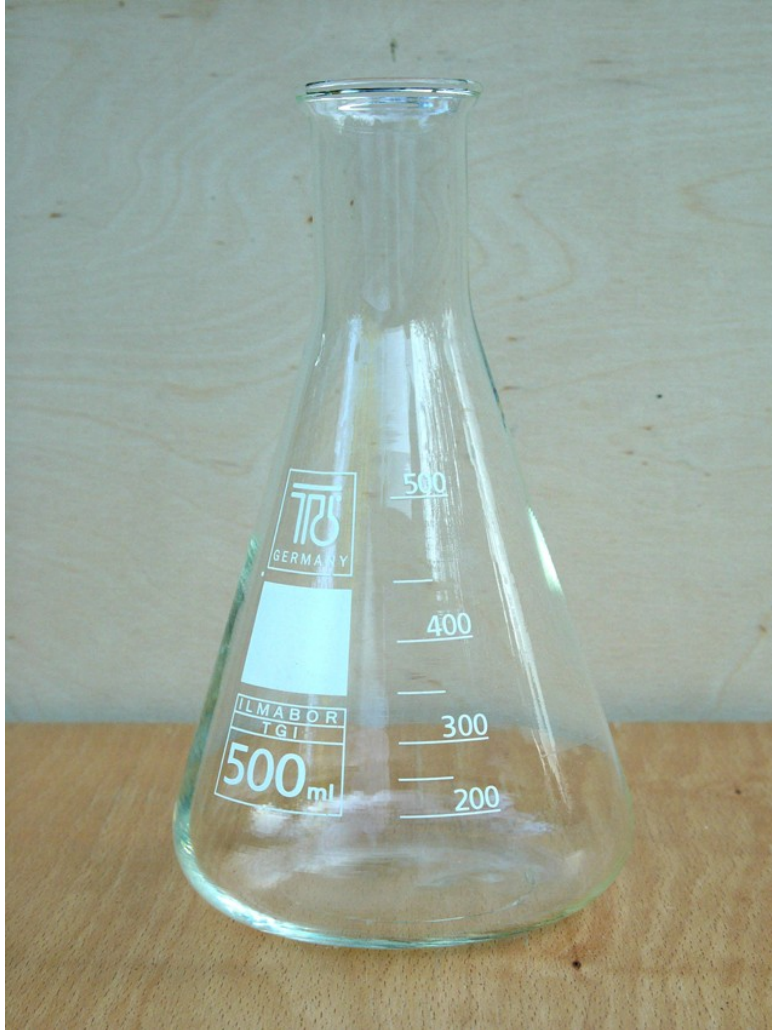
Properties of Electric Charges (4 of 4)



- Charge is quantized.

$$e = 1.602\,19 \times 10^{-19} \text{ C}$$

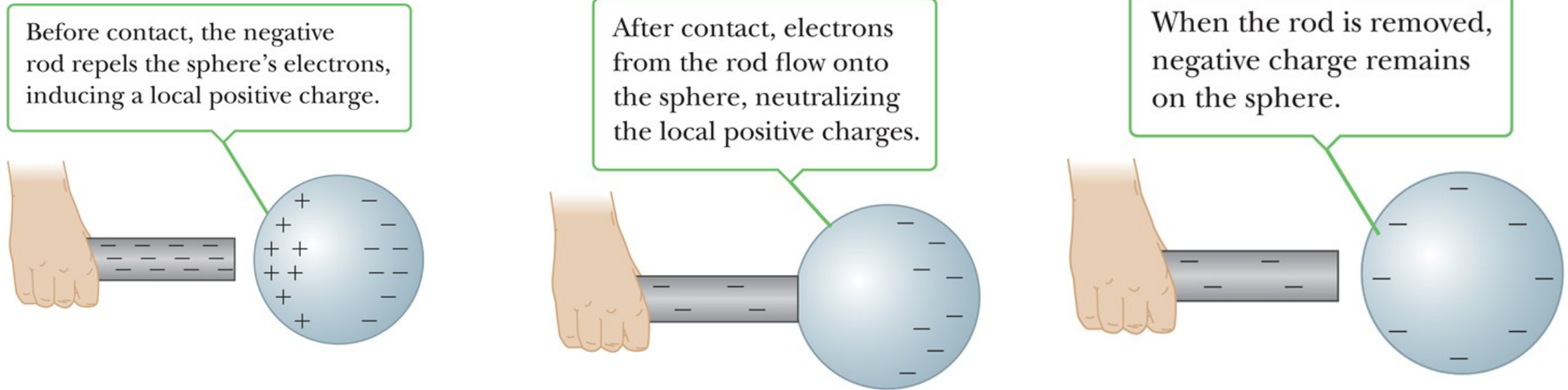
Insulators and Conductors (1 of 2)



Insulators and Conductors (2 of 2)

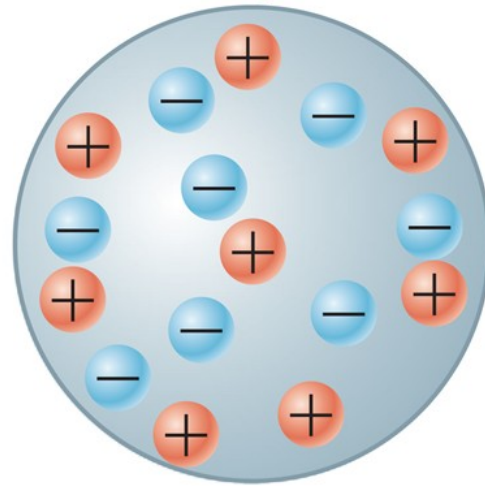


Charging by Conduction



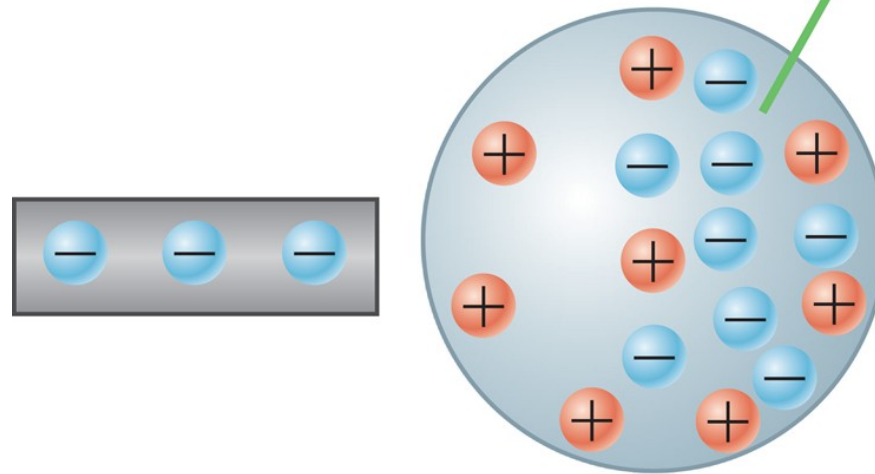
Charging by Induction (1 of 6)

The neutral sphere has equal numbers of positive and negative charges.

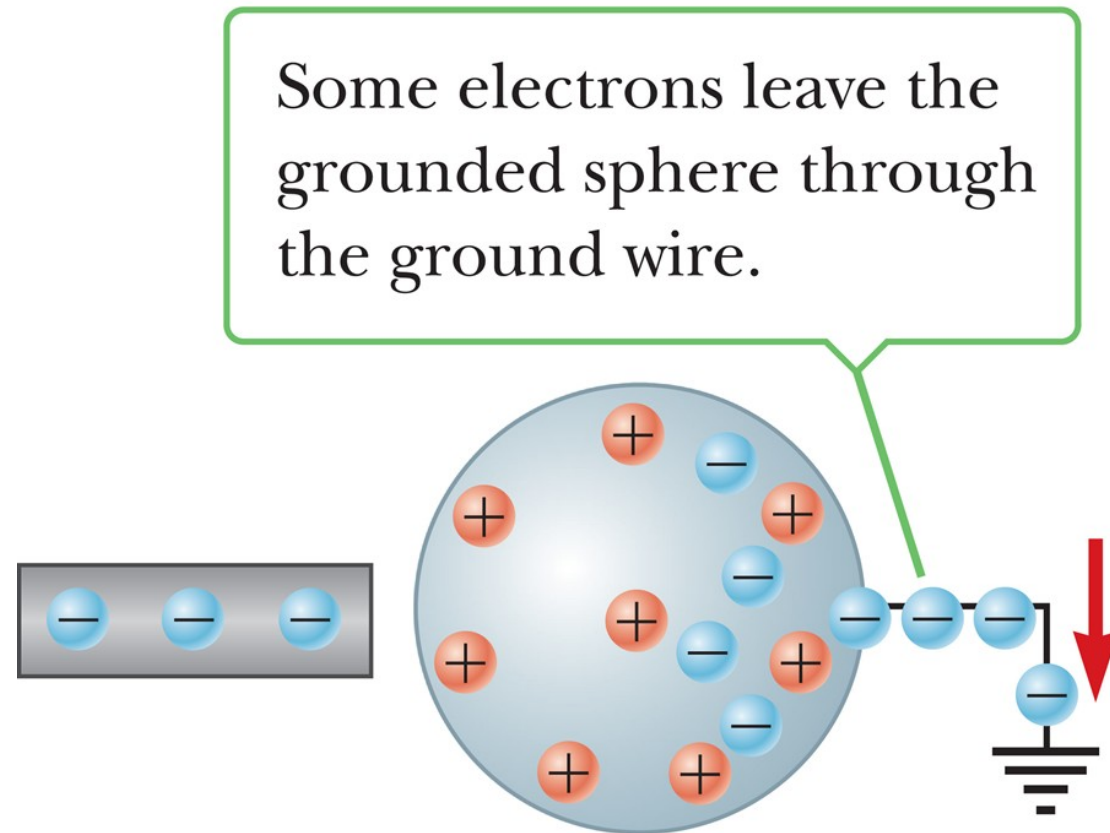


Charging by Induction (2 of 6)

Electrons redistribute when a charged rod is brought close.

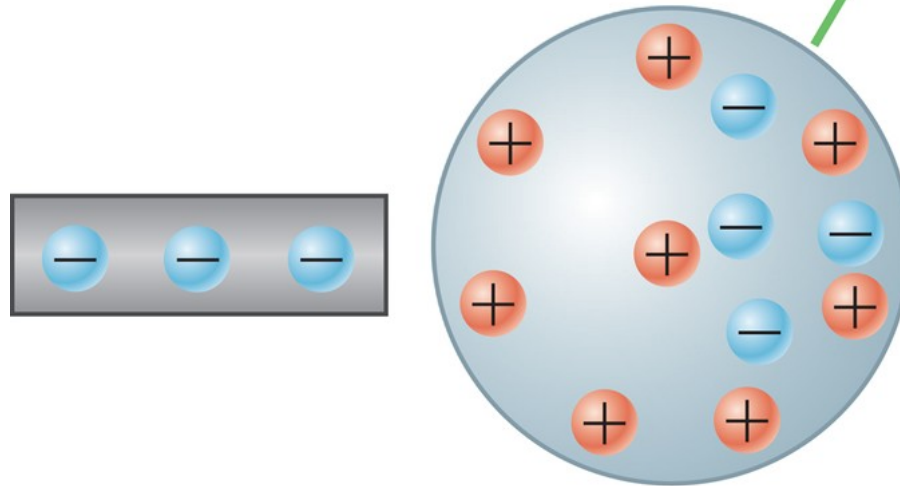


Charging by Induction (3 of 6)



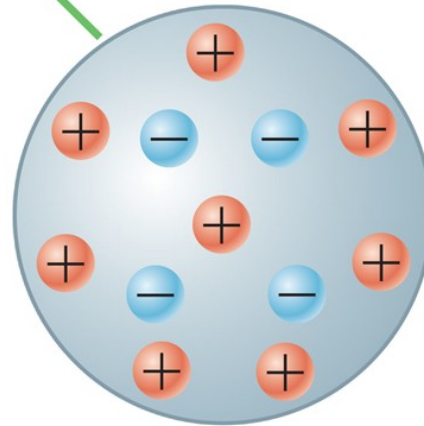
Charging by Induction (4 of 6)

The excess positive charge is nonuniformly distributed.

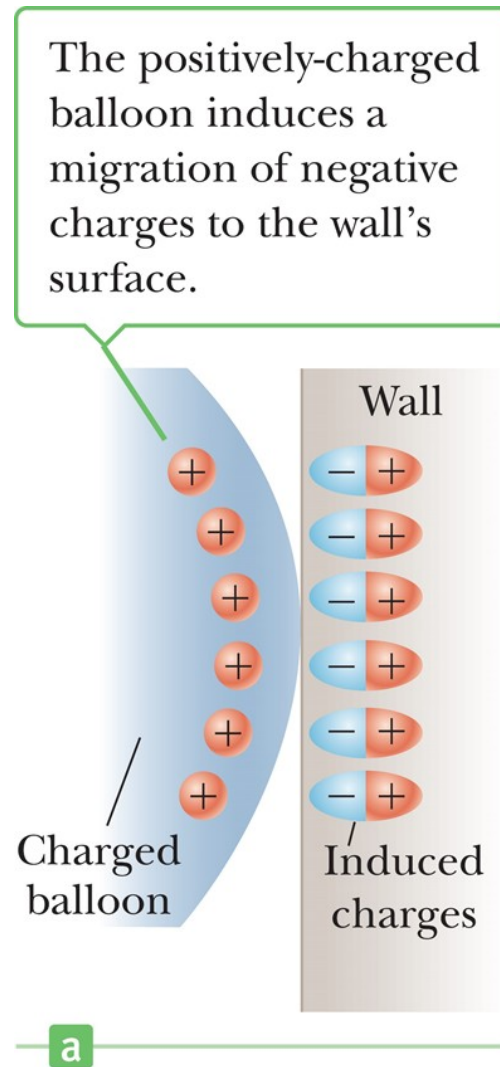


Charging by Induction (5 of 6)

The remaining electrons redistribute uniformly, and there is a net uniform distribution of positive charge on the sphere's surface.



Charging by Induction (6 of 6)



Think – Pair – Share 1

A suspended object A is attracted to a neutral wall. It's also attracted to a positively-charged object B. Which of the following is true about object A?

1. It is uncharged.
2. It has a negative charge.
3. It has a positive charge.
4. It may be either charged or uncharged.

Coulomb's Law (1 of 4)

Properties of Electric Force:

1. Directed along a line joining the two particles and inversely proportional to the square of the separation distance
2. Proportional to the product of the magnitudes of the charges of the two particles
3. Attractive if the charges are of opposite sign and repulsive if the charges have the same sign

$$F = k_e \frac{|q_1||q_2|}{r^2}$$

$$k_e = 8.987\,5 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

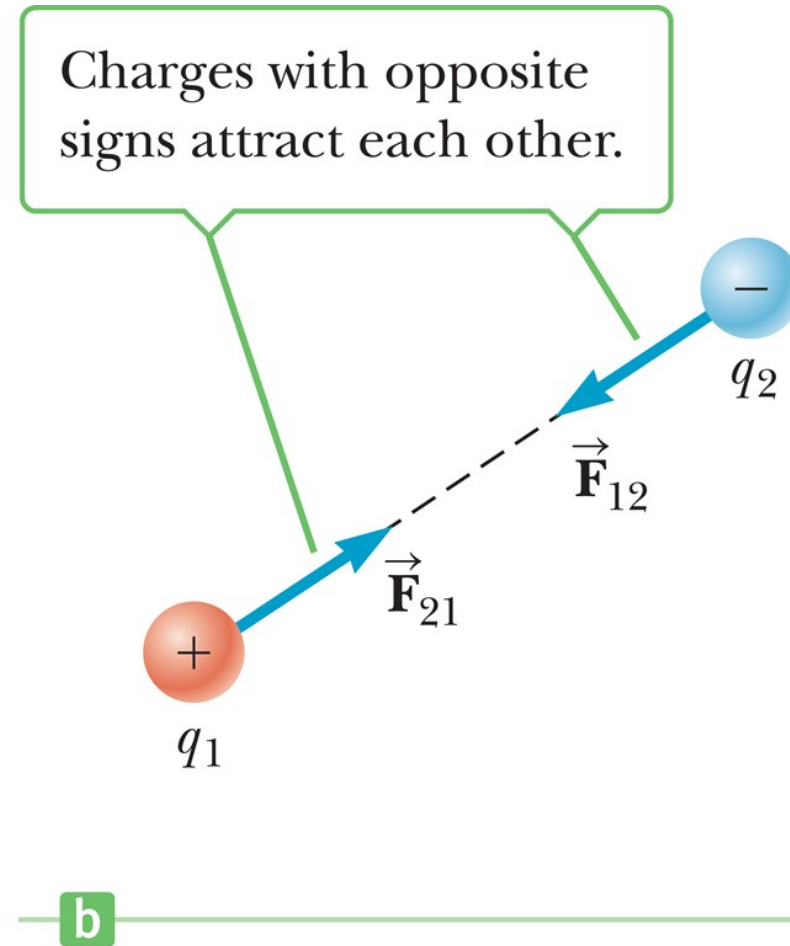
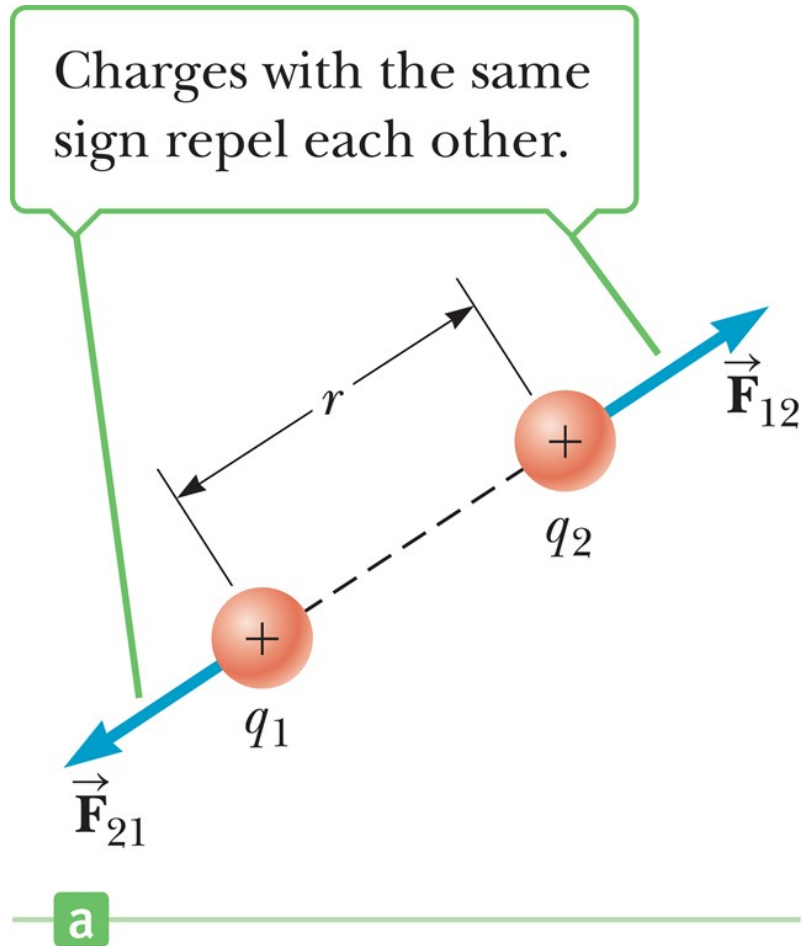
Coulomb's Law (2 of 4)

Table 15.1 Charge and Mass of the Electron, Proton, and Neutron

Particle	Charge (C)	Mass (kg)
Electron	-1.60×10^{-19}	9.11×10^{-31}
Proton	$+1.60 \times 10^{-19}$	1.67×10^{-27}
Neutron	0	1.67×10^{-27}

$$\frac{1}{e} = \frac{1 \text{ proton}}{1.6 \times 10^{-19} \text{ C}} = 6.3 \times 10^{18} \text{ protons/C}$$

Coulomb's Law (3 of 4)



Think – Pair – Share 2

Object A has a charge of $+2 \mu\text{C}$, and object B has a charge of $+6 \mu\text{C}$. Which statement is true?

1. $\vec{F}_{AB} = -3\vec{F}_{BA}$

2. $\vec{F}_{AB} = -\vec{F}_{BA}$

3. $3\vec{F}_{AB} = -\vec{F}_{BA}$

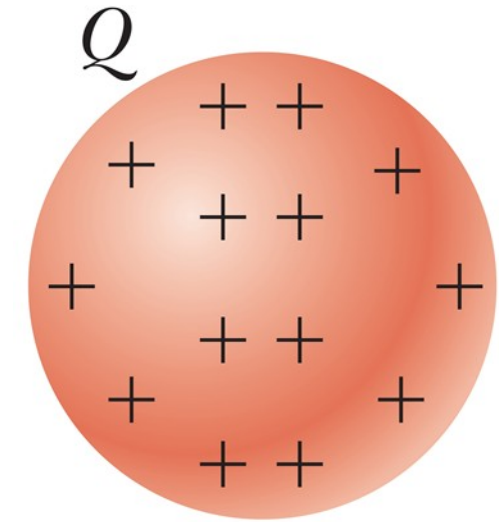
Coulomb's Law (4 of 4)

$$F = k_e \frac{|q_1||q_2|}{r^2}$$

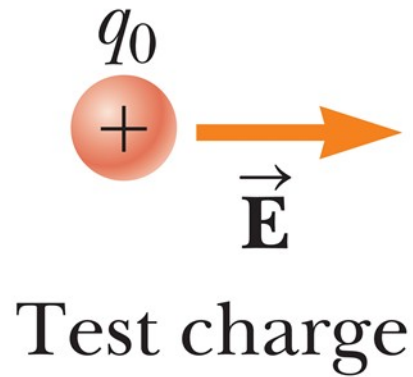
$$F_g = G \frac{m_1 m_2}{r^2}$$

1. Electric forces: attractive or repulsive; gravitational forces: always attractive
2. Electric force much stronger than gravitational force

Electric Fields (1 of 4)

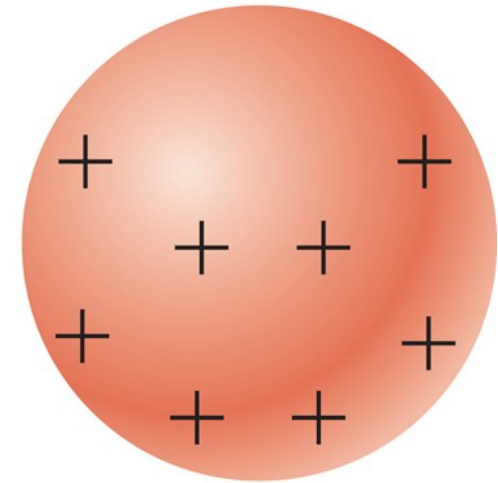
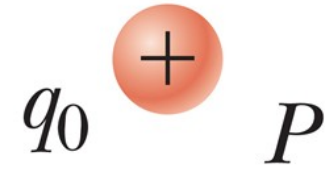


Source charge

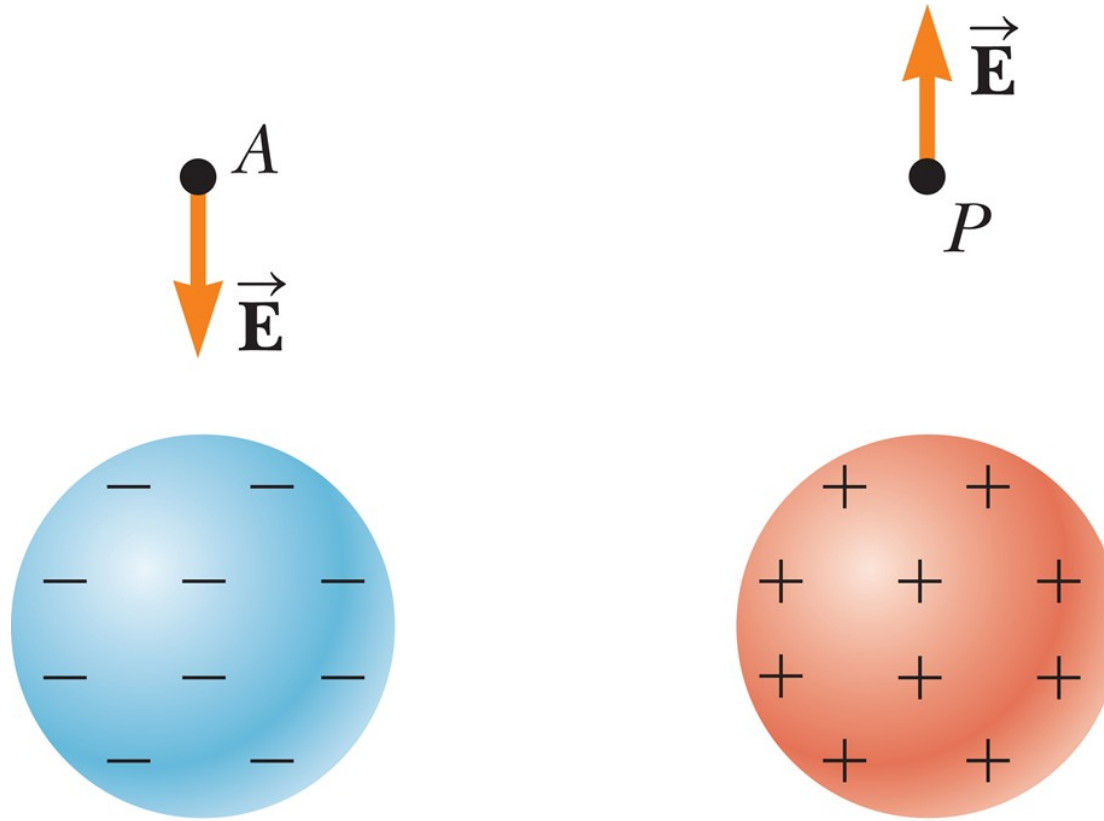


Test charge

$$\vec{E} \equiv \frac{\vec{F}}{q_0} \quad \text{SI unit: N/C}$$



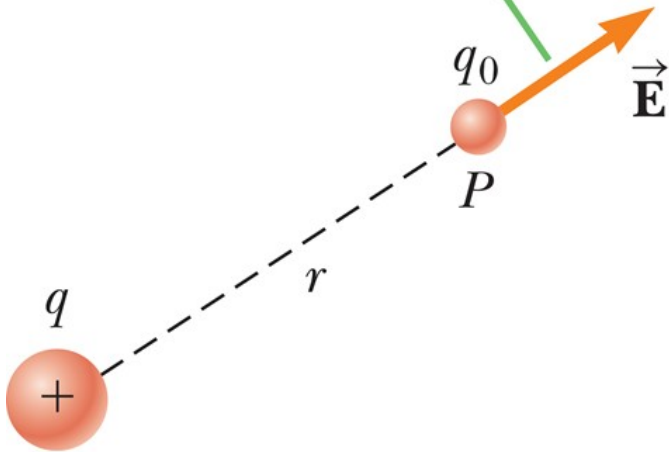
Electric Fields (2 of 4)



$$\vec{F} = q\vec{E}$$

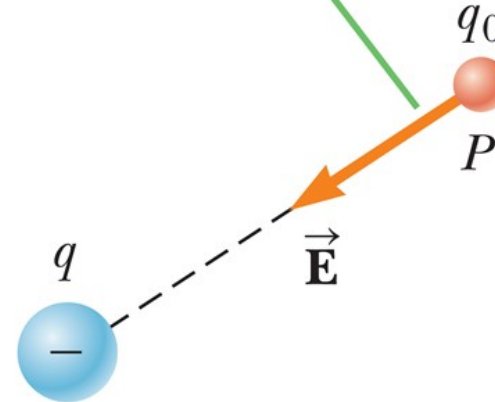
Electric Fields (3 of 4)

If q is positive, the electric field at P points radially outwards from q .



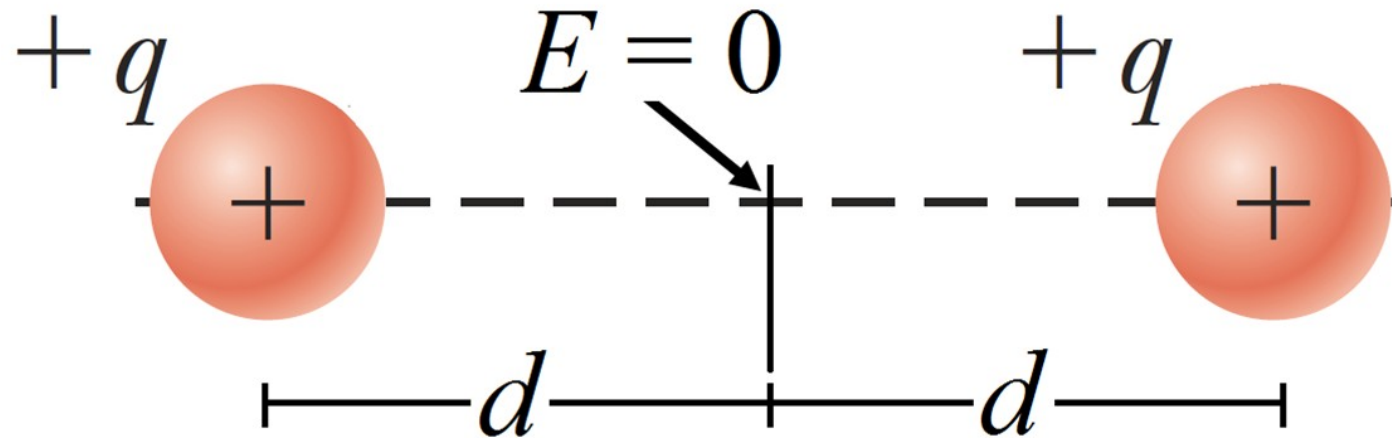
$$F = k_e \frac{|q| |q_0|}{r^2}$$

If q is negative, the electric field at P points radially inwards toward q .



$$E = k_e \frac{|q|}{r^2}$$

Electric Fields (4 of 4)



Think – Pair – Share 3

A test charge of $+3 \mu\text{C}$ is at a point P where the electric field due to other charges is directed to the right and has a magnitude of $4 \times 10^6 \text{ N/C}$.

If the test charge is replaced with a $-3 \mu\text{C}$ charge, the electric field at P

1. has the same magnitude as before, but changes direction.
2. increases in magnitude and changes direction.
3. remains the same.
4. decreases in magnitude and changes direction.

Think – Pair – Share 4

A circular ring of charge of radius b has a total charge q uniformly distributed around it. Choose the magnitude of the electric field at the center of the ring.

1. 0
2. $k_e q / b^2$
3. $k_e q^2 / b^2$
4. $k_e q^2 / b$
5. None of these answer choices is correct.

Think – Pair – Share 5

A “free” electron and a “free” proton are placed in an identical electric field. Is the following statement true or false?

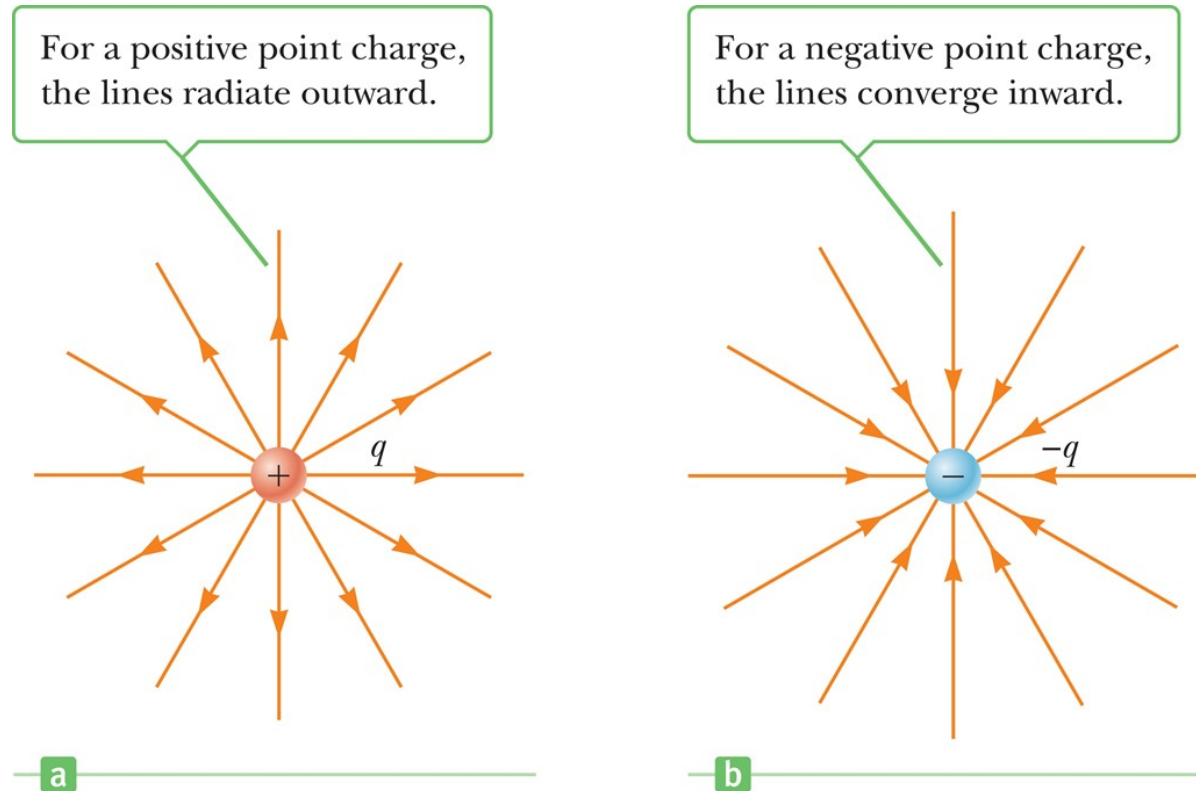
Each particle is acted upon by equal-magnitude electric forces and both have the same magnitude of acceleration.

1. False
2. True

Problem-Solving Strategy: Calculating Electric Forces and Fields

1. Draw
2. Identify charge
3. Convert all units
4. Apply Coulomb's law
5. Sum all the x-components
6. Sum all the y-components
7. Use the Pythagorean theorem and trigonometry

Electric Field Lines (1 of 4)



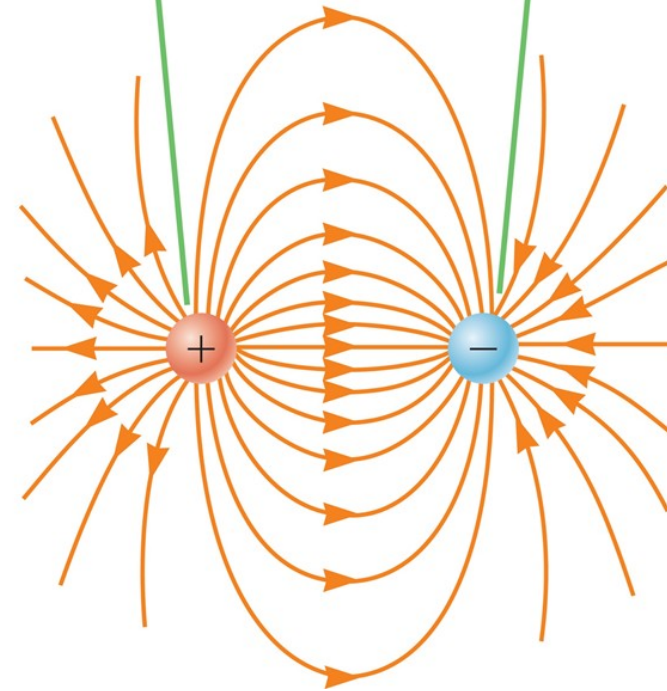
1. $\mathbf{E}$ is tangent to the electric field lines at each point
2. The number of lines proportional to the field strength

Electric Field Lines (2 of 4)

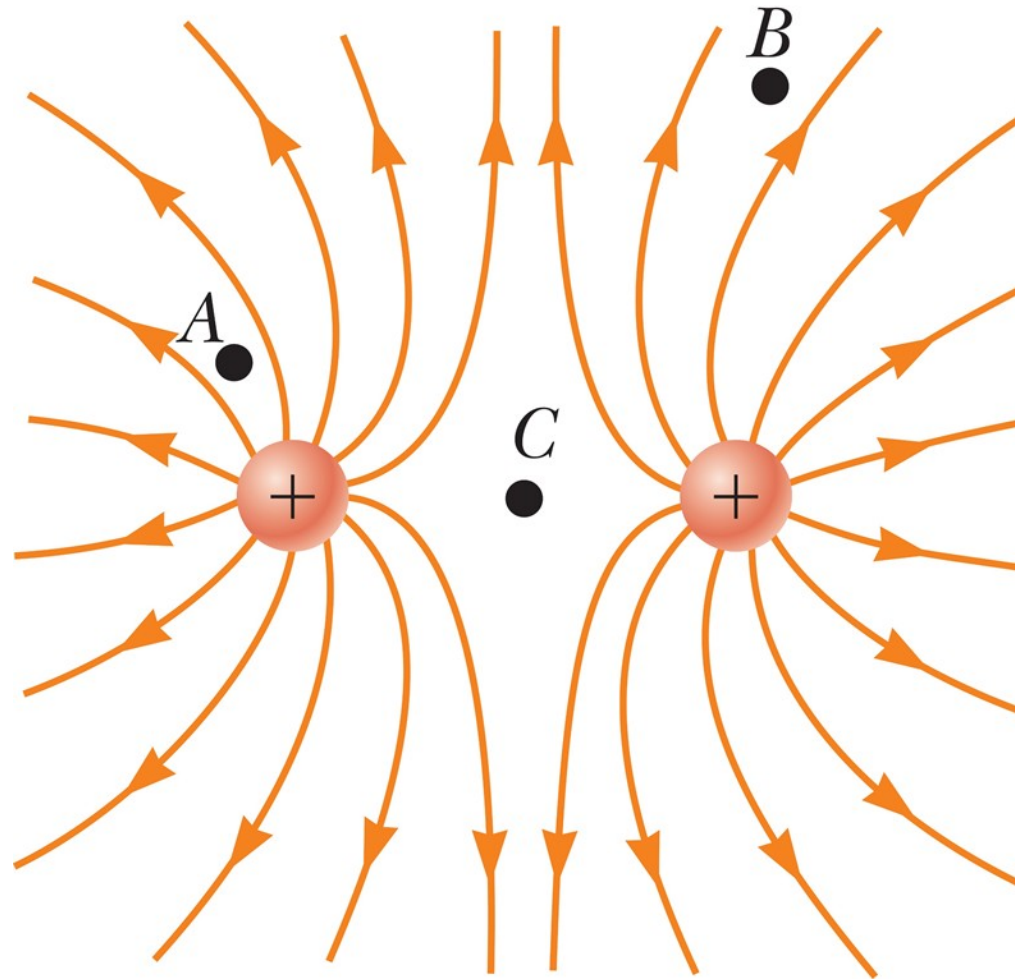
Drawing Field Lines:

1. Lines begin on positive charges and end on negative charges
2. Number of lines is proportional to charge magnitude
3. Field lines cannot cross

The number of field lines leaving the positive charge equals the number terminating at the negative charge.

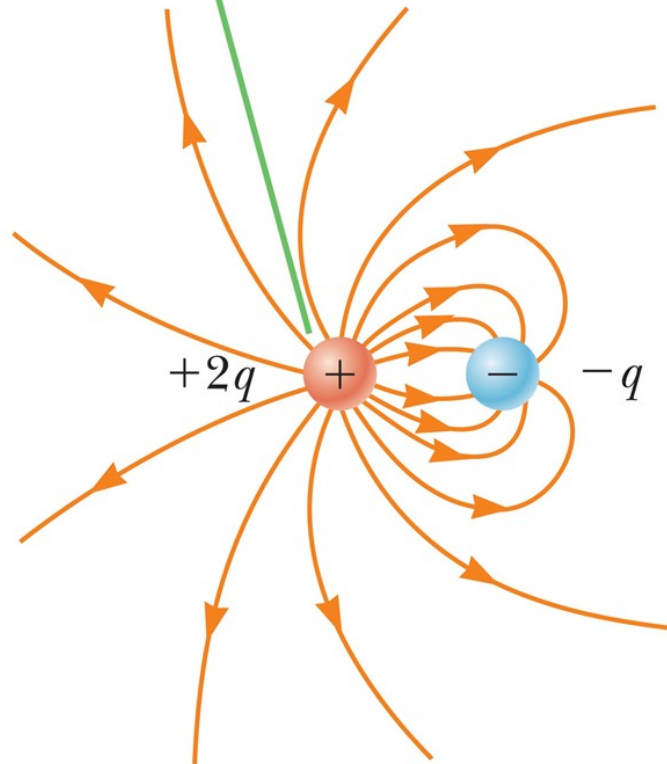


Electric Field Lines (3 of 4)



Electric Field Lines (4 of 4)

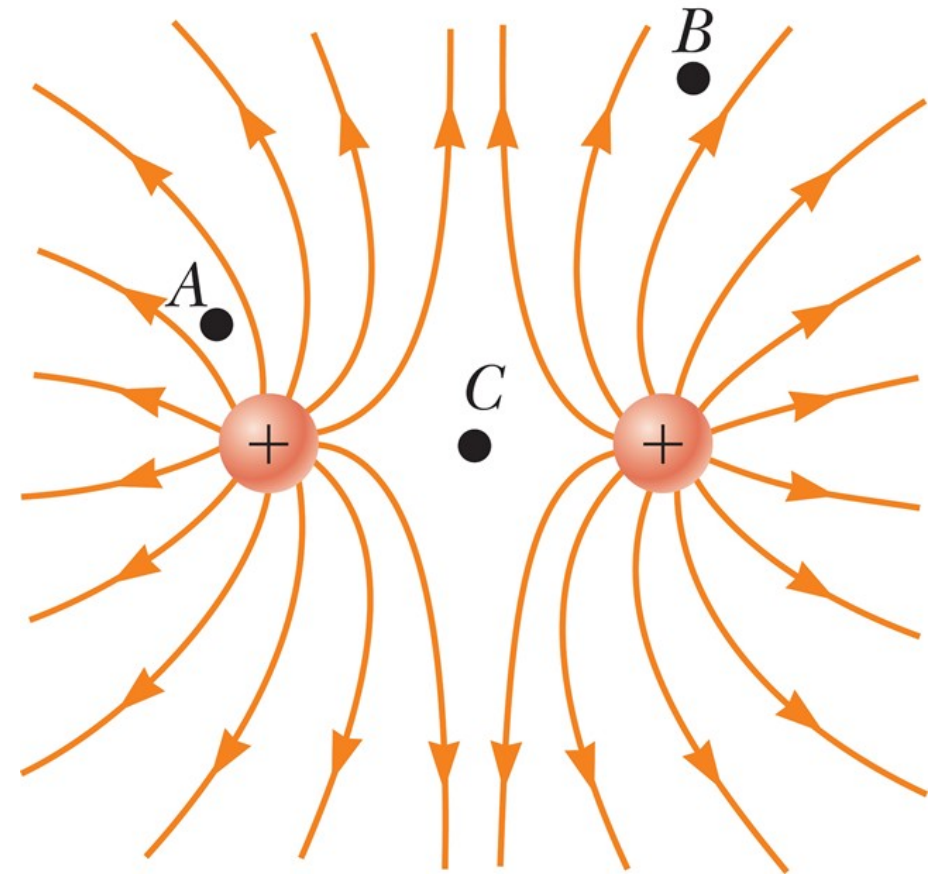
Two field lines leave $+2q$ for every one that terminates on $-q$.



Think – Pair – Share 6

Rank the magnitude of the electric field at points A , B , and C in the figure below, largest magnitude first.

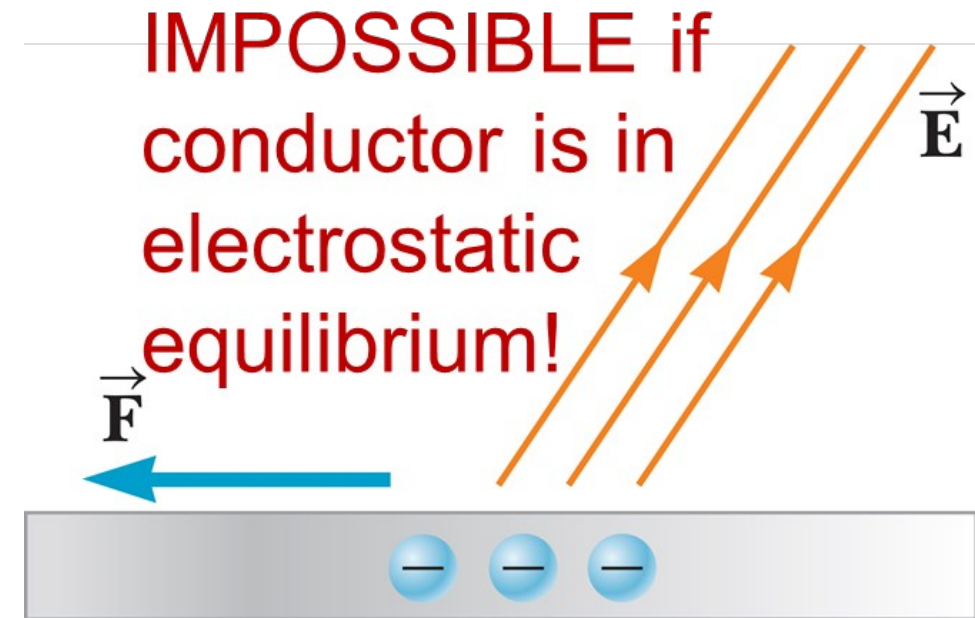
1. A , B , C
2. A , C , B
3. C , A , B
4. The answer can't be determined by visual inspection.



Conductors in Electrostatic Equilibrium (1 of 4)

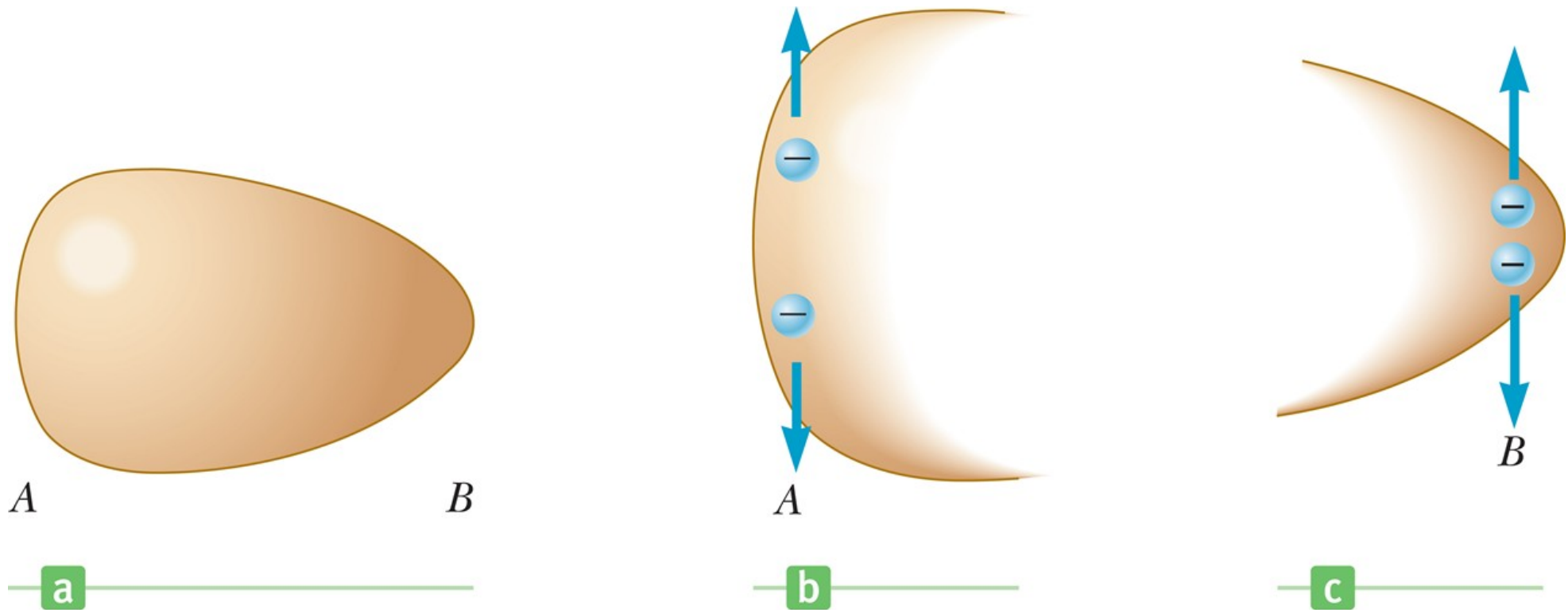
Properties of Electrostatic Equilibrium

1. The electric field is zero everywhere inside the conducting material.
2. Any excess charge on an isolated conductor resides entirely on its surface.
3. The electric field just outside a charged conductor is perpendicular to the conductor's surface.

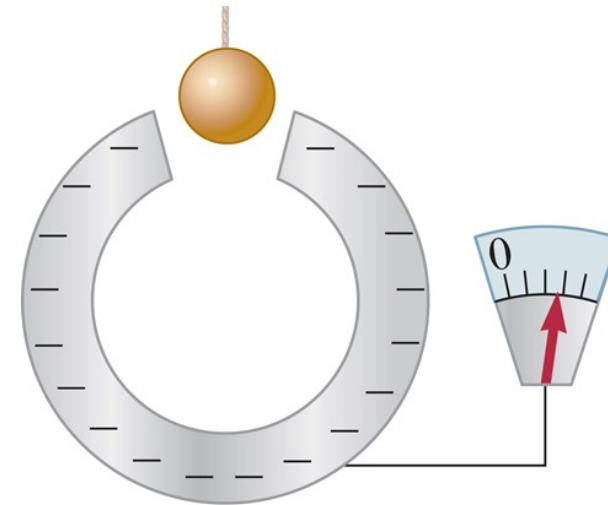
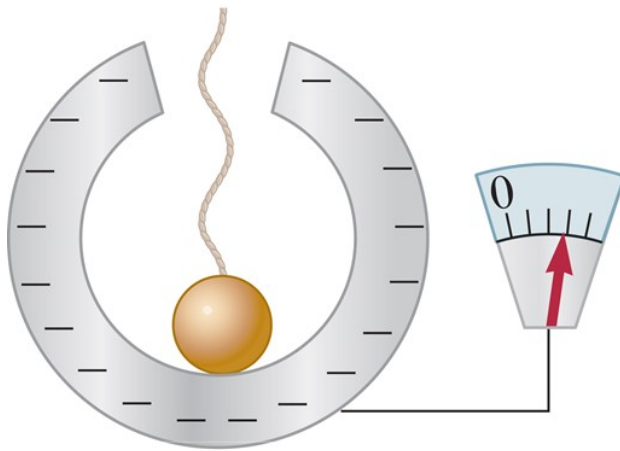
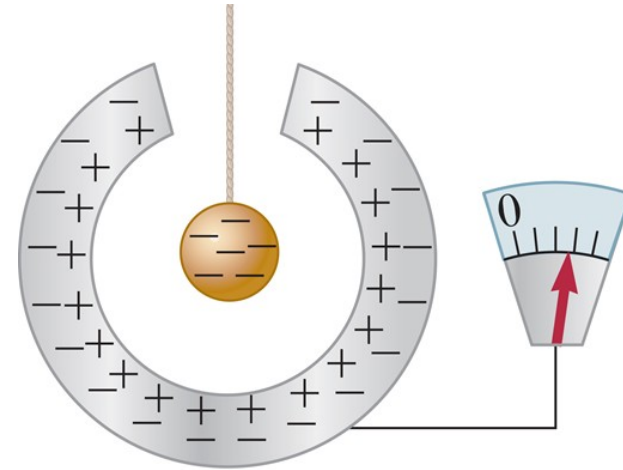
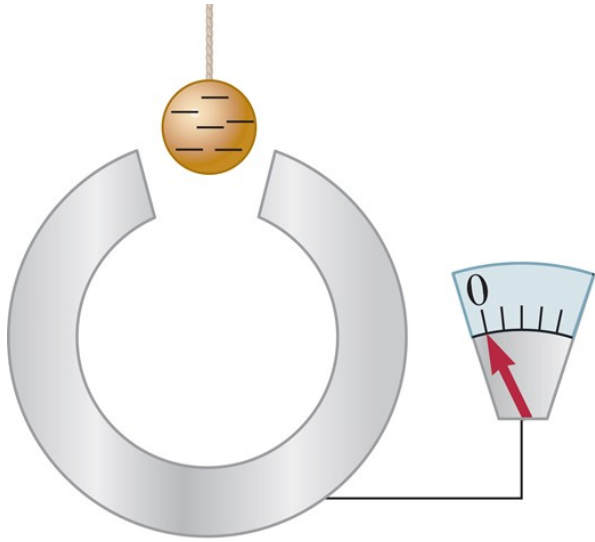


Conductors in Electrostatic Equilibrium (2 of 4)

4. On an irregularly shaped conductor, the charge accumulates at sharp points, where the radius of curvature of the surface is smallest.



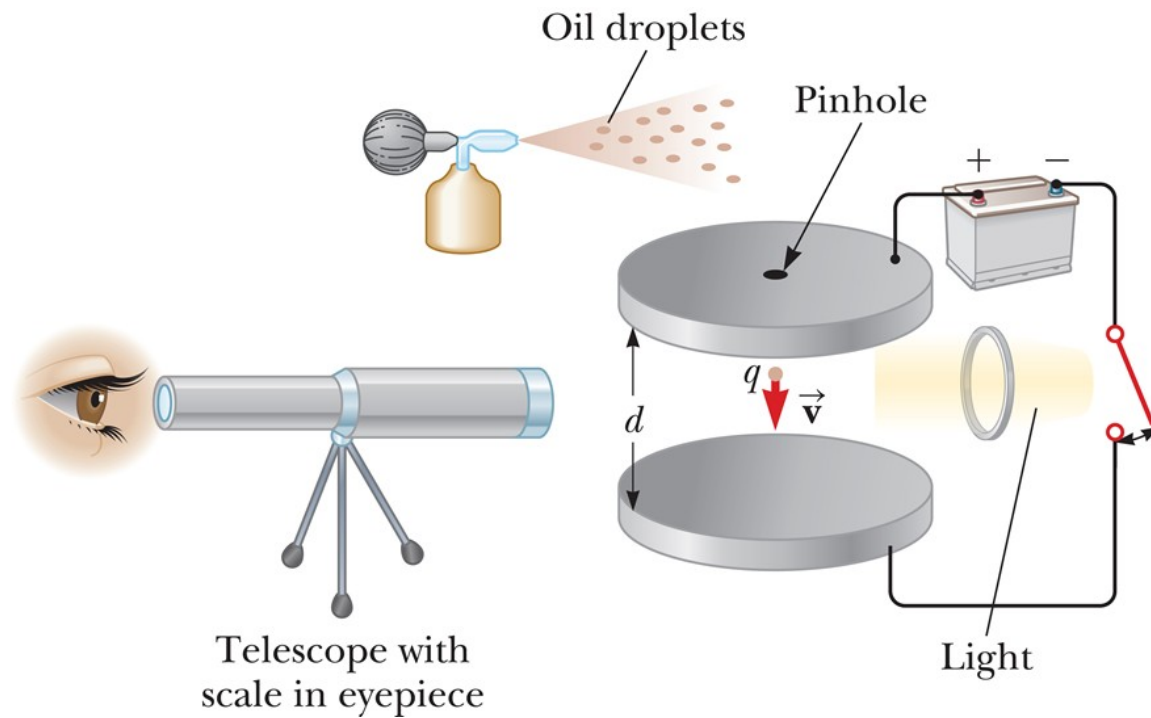
Conductors in Electrostatic Equilibrium (3 of 4)



Conductors in Electrostatic Equilibrium (4 of 4)



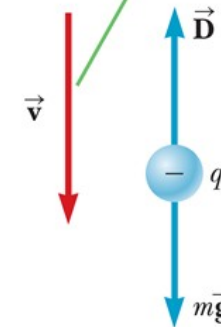
The Millikan Oil-Drop Experiment



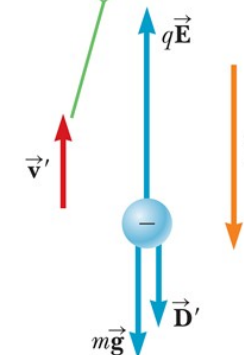
$$q = ne \quad (n = 0, \pm 1, \pm 2, \pm 3, \dots)$$

$$e = 1.60 \times 10^{-19} \text{ C}$$

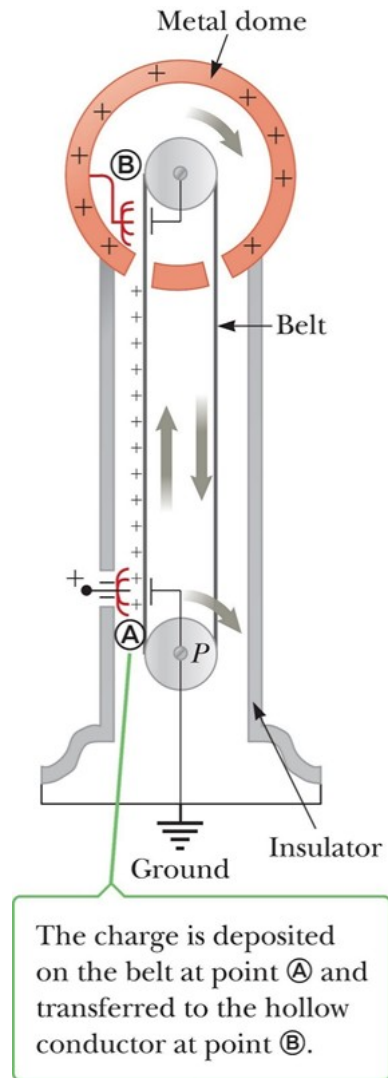
Electric field off: the droplet falls at terminal velocity $\vec{v}$, the gravity and drag forces summing to zero.



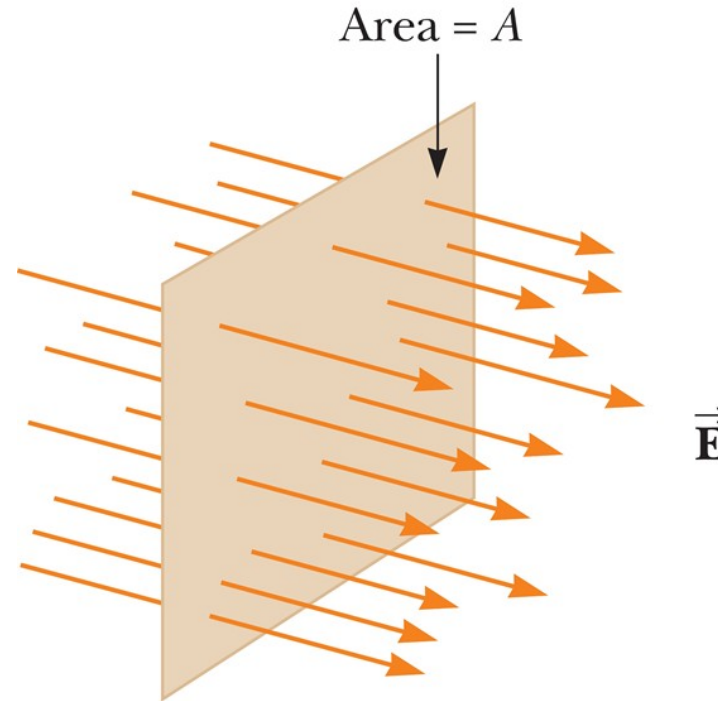
Electric field on: the droplet moves upward at new terminal velocity $\vec{v}'$, the gravity, drag, and electric forces summing to zero.



The Van de Graaff Generator



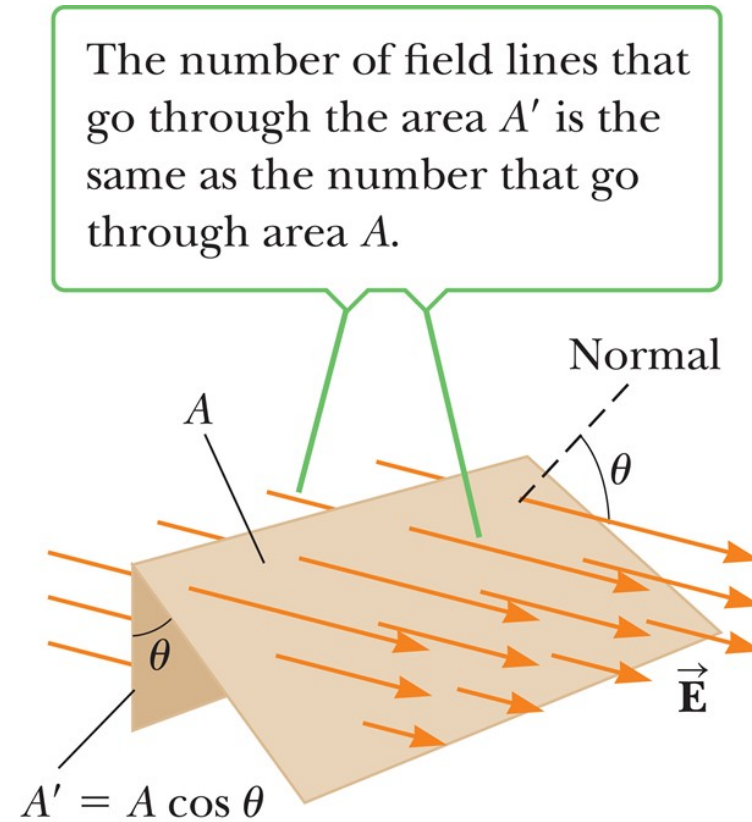
Electric Flux and Gauss's Law



$$\frac{N}{A} \propto E \quad \rightarrow \quad N \propto EA$$

$$\Phi_E = EA \quad \text{SI unit: } \text{N} \cdot \text{m}^2/\text{C}$$

Electric Flux



$$\Phi_E = EA \cos \theta$$

Think – Pair – Share 7

Calculate the magnitude of the flux of a constant electric field of 5.00 N/C in the z -direction, through a rectangle with area 4.00 m^2 in the xy -plane.

1. 0
2. $10.0 \text{ N}\cdot\text{m}^2/\text{C}$
3. $20.0 \text{ N}\cdot\text{m}^2/\text{C}$
4. More information is needed.

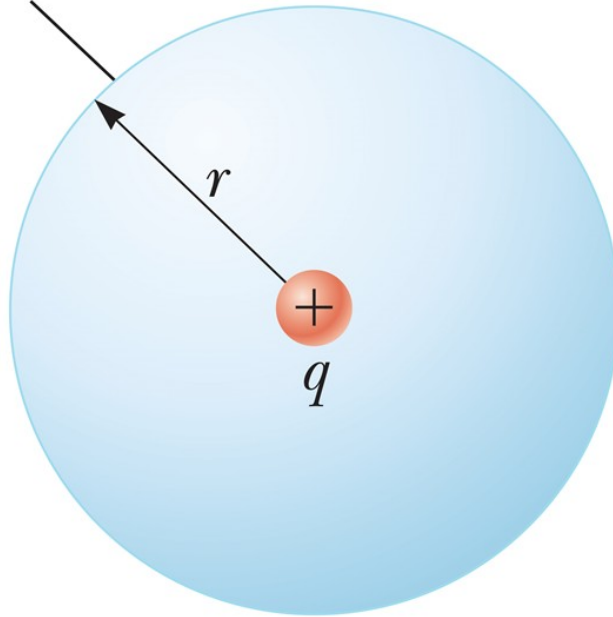
Think – Pair – Share 8

Calculate the magnitude of the flux of a constant electric field of 5.00 N/C tilted 60° away from the positive z -direction, through a rectangle with area 4.00 m^2 in the xy -plane.

1. 0
2. $10.0 \text{ N}\cdot\text{m}^2/\text{C}$
3. $20.0 \text{ N}\cdot\text{m}^2/\text{C}$
4. More information is needed.

Gauss's Law (1 of 3)

Gaussian
surface



$$E = k_e \frac{q}{r^2}$$

$$\Phi_E = EA$$

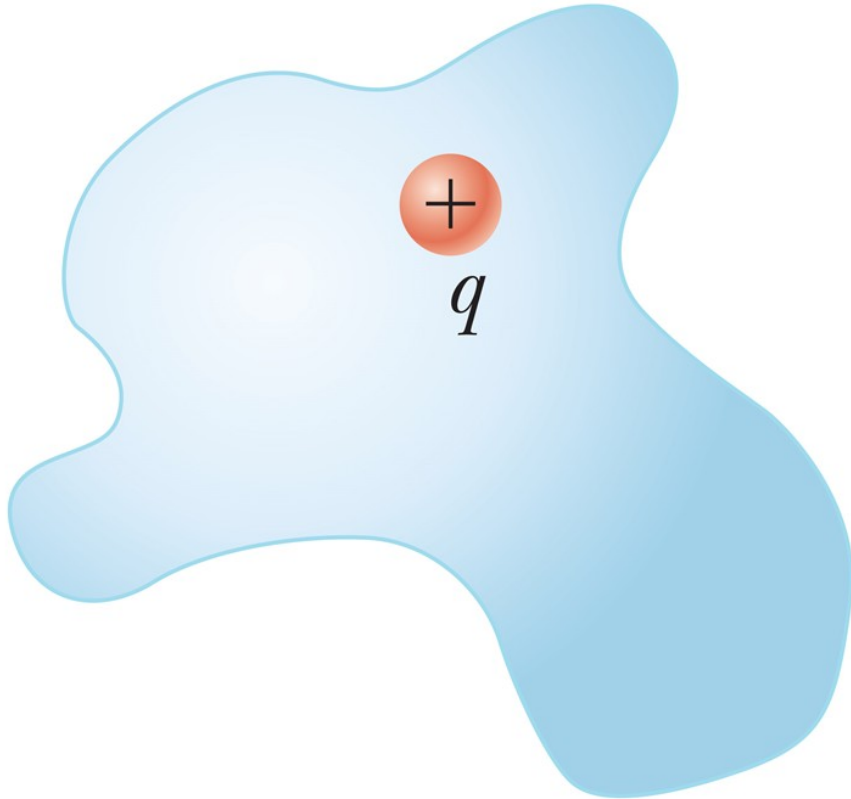
$$= k_e \frac{q}{r^2 (4\pi r^2)}$$

$$= 4\pi k_e q$$

$$k_e = \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 = \frac{1}{4\pi k_e} = 8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2)$$

Gauss's Law (2 of 3)



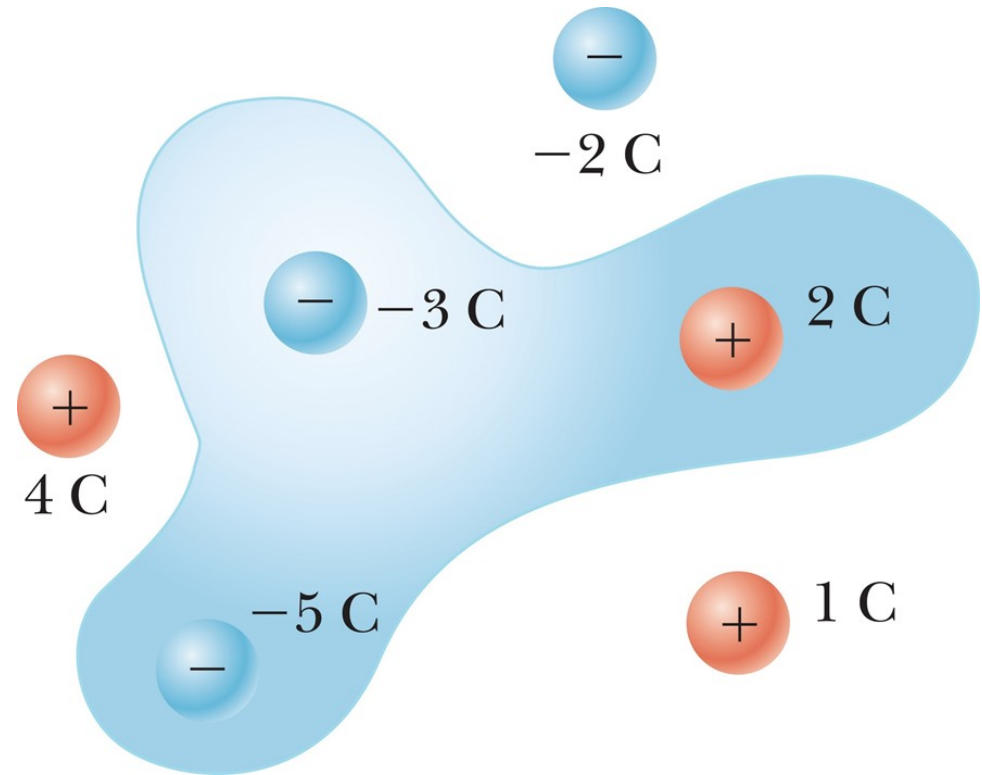
$$\Phi_E = 4 \pi k_e q = \frac{q}{\epsilon_0}$$

$$\Phi_E = \frac{Q_{\text{inside}}}{\epsilon_0}$$

Think – Pair – Share 9

Find the electric flux through the surface in the figure below.

1. $-(3 \text{ C})/\epsilon_0$
2. $(3 \text{ C})/\epsilon_0$
3. 0
4. $-(6 \text{ C})/\epsilon_0$



Think – Pair – Share 10

For a closed surface through which the net flux is zero, each of the following four statements *could* be true. Which of the statements *must* be true? (There may be more than one.)

- (a) There are no charges inside the surface.
- (b) The net charge inside the surface is zero.
- (c) The electric field is zero everywhere on the surface.
- (d) The number of electric field lines entering the surface equals the number leaving the surface.

1. (a) only
2. (b) only
3. (b) and (d)
4. (a) and (d)

Gauss's Law (3 of 3)

